Machine Learning and IoT Based Traffic Management System

Prioritizing Emergency Vehicles and Reducing Congestion

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Outline

- **Problem Statement**
- **Research Objectives**
- **Methodology**
- **Results**
- **Implementation Challenges**
- **Economic Impact**
- **Conclusions**
- **Future Work**

Traffic Crisis in Dhaka

- **Severe Congestion**: Average vehicle speed drops to 4.8 km/h
- **Economic Impact**: $3.8 billion annual cost
- **Emergency Delays**: 56% of emergency responses delayed
- **Infrastructure Limits**: Roads designed for smaller population
- **Mixed Traffic**: Complex vehicle types competing for space

Figure: Shahbag Traffic Congestion

Need for Intelligent Traffic Management

Current Traffic Management Challenges

Technical Challenges:

- Fixed timing signals
- No real-time adaptation
- Manual traffic control
- Limited sensor integration
- Poor emergency prioritization

Impact on Society:

- 22 million population affected
- Daily productivity losses
- Increased fuel consumption
- Emergency service delays

Figure: Traffic Congestion at Major Intersections

Traditional systems cannot handle dynamic traffic patterns

Research Motivation

1. **Urbanization Crisis**
   - Population growth exceeds infrastructure development
   - Vehicle ownership increasing exponentially
2. **Technology Opportunity**
   - IoT and ML technologies are now mature
   - Real-time processing capabilities available
3. **Life-Critical Need**
   - Emergency services facing critical delays
   - Traffic accidents on the rise
4. **Economic Necessity**
   - Massive economic losses due to traffic
   - Cost-effective solutions needed

Figure: Traffic Scenario in Narayanganj

Key Objectives

1. **Real-time Vehicle Detection**
   - YOLOv11 based object detection
   - Classify vehicles, pedestrians, emergency vehicles
2. **Emergency Vehicle Prioritization**
   - Automatic detection and signal adjustment
   - Reduce emergency response times
3. **Adaptive Traffic Control**
   - Dynamic signal timing based on traffic density
   - Prevent lane starvation
4. **IoT Integration**
   - Arduino, Raspberry Pi, NodeMCU
   - Connect ML algorithms with physical infrastructure

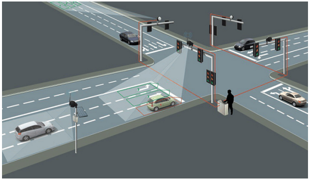
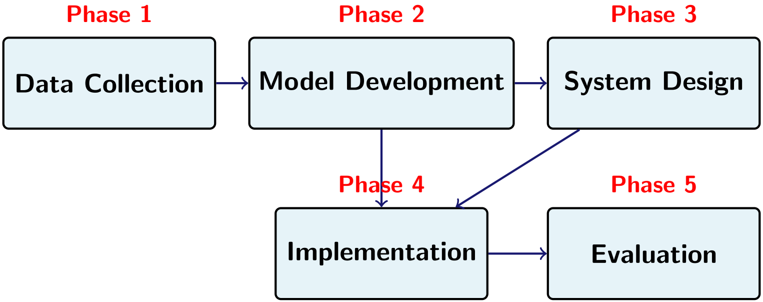


Figure: CCTV Infrastructure

Research Methodology Overview



5-Phase Systematic Approach:

- Phase 1: Comprehensive traffic data collection from 7 strategic locations
- Phase 2: YOLOv11 model training and optimization
- Phase 3: IoT-based system architecture design
- Phase 4: Real-world pilot deployment
- Phase 5: Performance evaluation and analysis

Data Collection Strategy

Strategic Locations:

1. Shahbag (Academic area)
2. Polton (Residential/Commercial)
3. Motijheel (Business district)
4. Science Lab (Transit hub)
5. Panthapath (Major arterial)
6. Bijoy Sarani (North-South corridor)
7. Gulistan (Dense urban area)

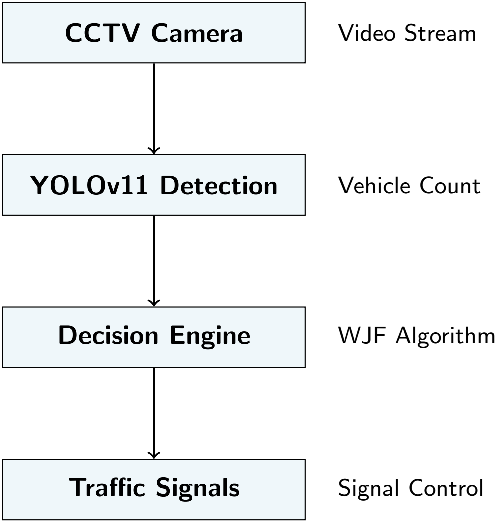
Data Characteristics:

- 6-month collection period
- Multiple time periods
- Various weather conditions
- High-resolution imagery
- Diverse traffic scenarios

Quality Assurance:

- Multi-level annotation review
- Expert domain validation
- Consistency checks
- Data augmentation techniques

System Architecture



YOLOv11 Model Architecture

Why YOLOv11?

- Real-time processing capability
- Superior accuracy vs previous versions
- Efficient memory usage
- Multi-scale object detection
- Robust performance in challenging conditions

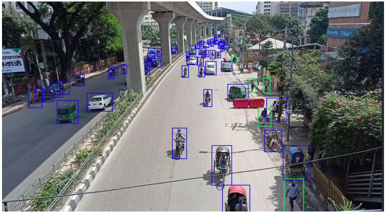


Figure: YOLOv11 Detection Example

Model Specifications:

- Input: 640x640 pixels
- Backbone: CSPDarkNet53
- Neck: PANet
- Head: YOLOv11 Detection Head

Training Details:

Parameter	Value
Model	YOLOv11m
Epochs	256
Batch Size	16
Learning Rate	0.01
Optimizer	AdamW

Dataset and Training

Dataset Statistics:

- 3,784 images collected
- 171,436 annotated objects
- 7 strategic locations
- 21 vehicle categories

Training Configuration:

- YOLOv11m model
- 256 epochs
- 640x640 resolution

Algorithm: Real-Time Emergency Vehicle Detection and Traffic Management

1. Start
2. Start analyzing real-time CCTV footage from all lanes.
3. For each lane, detect:
   - Number of Vehicles
   - Number of Emergencies Vehicles
   - Number of Pedestrians
4. If an emergency vehicle is detected:
   - Immediately open the corresponding lane.
   - If another emergency vehicle is detected in a different lane:
     - Wait until the first emergency lane is cleared.
5. For non-emergency scenarios:
   - Detect the number of vehicles and persons for each lane.
   - Calculate lane priority using the Weighted Job First algorithm.
   - Consider avoiding long starvation for any lane by ensuring:
     - Each lane stays open for a minimum time limit once opened.
     - No lane exceeds its maximum time limit.
6. Open the lane with the highest priority based on the calculated weights.
7. To manage traffic lights (green/yellow/red), send control signals to the hardware (e.g., Arduino, Raspberry Pi, NodeMCU).
8. Continue the process in a loop to maintain traffic signal control.
9. Stop

Model Performance Overview

Detection Accuracy:

- **Overall mAP50**: 79%
- **Vehicle Detection**: 85%
- **Pedestrian Detection**: 82%
- **Emergency Vehicle**: 85%

System Performance:

- **Response Time**: 222ms avg
- **System Uptime**: 98.9%
- **Processing Rate**: 30+ FPS
- **Memory Usage**: 2.1GB

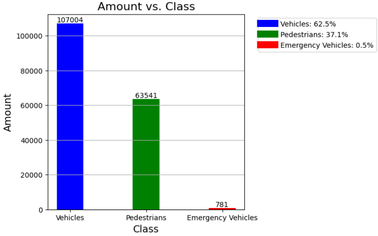
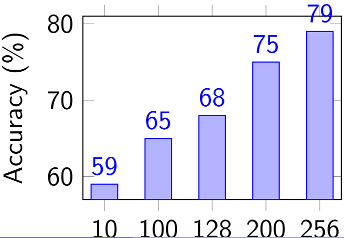


Figure: Performance Metrics



Confusion Matrix Analysis

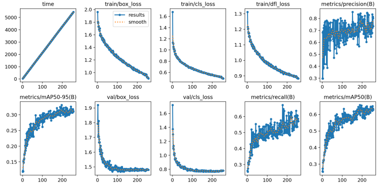


Figure: Confusion Matrix Visualization

Table: Model Confusion Matrix

True/Pred	Emerg	Person	Vehicle	BG
Emergency	**4**	3	0	1
Person	0	**2080**	47	890
Vehicle	0	41	**4111**	1191
Background	0	1152	0	**1222**

Performance Metrics:

- **Precision**: 83.2%
- **Recall**: 81.7%
- **F1-Score**: 82.4%
- **Training Time**: 45.6 hours

Performance Improvements

Table: Traffic Flow Improvements

Metric	Traditional	Proposed	Improvement
Wait Time	12.5 min	8.4 min	**33% ↓**
Emergency	8.5 min	3.7 min	**56% ↓**
Lane Util.	70%	92%	**31% ↑**
Starvation	3.2/hr	0.1/hr	**97% ↓**

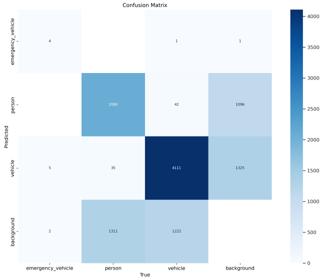


Figure: Performance Comparison

Significant improvements in all key metrics

Real-World Deployment Results

Pilot Locations:

1. Shahbag Intersection
2. Motijheel Commercial Area
3. Science Lab Junction

Deployment Duration:

- 2-month pilot study
- 24/7 continuous operation
- Multiple weather conditions
- Peak & off-peak hours

User Feedback:

- 87% overall satisfaction
- 92% noticed traffic improvements
- 79% support system expansion

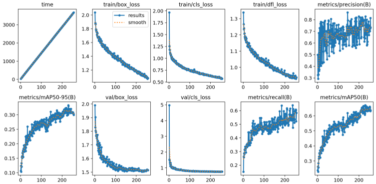


Figure: Deployment Results

System Reliability:

Location	Uptime	Response
Shahbag	98.7%	245ms
Motijheel	99.2%	198ms
Science Lab	98.9%	223ms
Average	**98.9%**	**222ms**

Technical Challenges & Solutions

Data Challenges:

- Challenge: Class imbalance (0.5% emergency vehicles)
- Solution: Data augmentation & weighted training

Hardware Limitations:

- Challenge: Processing power constraints
- Solution: Model optimization & edge computing

Environmental Factors:

- Challenge: Weather & lighting variations
- Solution: Robust training dataset & preprocessing

Integration Issues:

- Challenge: Legacy traffic infrastructure
- Solution: Modular IoT architecture

Real-time Requirements:

- Challenge: Sub-second response needed
- Solution: Optimized algorithms & parallel processing

Scalability Concerns:

- Challenge: City-wide deployment costs
- Solution: Cost-effective hardware & phased rollout

Cost-Benefit Analysis

- **Annual Net Benefits**: $8.065 million
- **Return on Investment**: 672%
- **CO2 Reduction**: 1,245 tons annually
- **Fuel Savings**: 485,000 liters annually
- **User Satisfaction**: 87% overall satisfaction

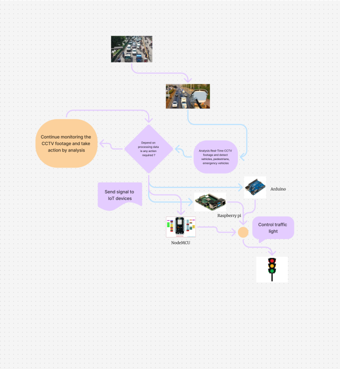
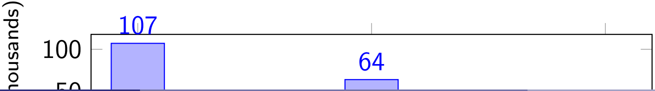


Figure: Economic Impact Analysis



Key Contributions

1. **Novel YOLOv11 Application**
   - First comprehensive application for Dhaka traffic
   - 79% detection accuracy achieved
2. **Emergency Vehicle Prioritization**
   - 56% reduction in response times
   - Automated detection and control
3. **Adaptive Traffic Management**
   - 33% reduction in wait times
   - 97% reduction in lane starvation
4. **Cost-Effective Solution**
   - 672% return on investment
   - Commercial hardware components

Future Research Directions

Short-term (6-12 months):

- Emergency vehicle dataset expansion
- Weather-resistant algorithms
- Hardware optimization
- User interface improvements
- Security enhancements

Medium-term (1-2 years):

- Predictive traffic modeling
- V2I communication integration
- Advanced ML algorithms
- Mobile app development
- Real-time analytics dashboard

Long-term (2-5 years):

- City-wide deployment
- Smart city ecosystem integration
- Autonomous vehicle compatibility
- International standardization
- AI-driven traffic optimization

Research Opportunities:

- Federated learning implementation
- Edge computing optimization
- Behavioral pattern analysis
- Environmental impact modeling
- Policy framework development

Thank You!

Questions & Discussion

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